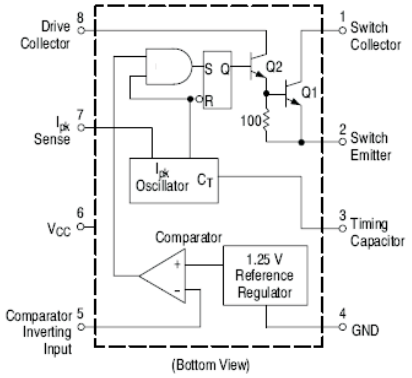
1.2 A, Step-Up/Down/Inverting Switching Regulators

MC34063

SOP8

■ Features

- Low Standby Current
- Current Limiting
- Output Switch Current to 1.2 A
- Output Voltage Adjustable
- Frequency Operation to 100 kHz
- Precision 2% Reference



■ Absolute Maximum Ratings Ta = 25℃

Parameter	Symbol	Rating	Unit
Power Supply Voltage	V_{CC}	40	V
Comparator Input Voltage Range	V_{IR}	-0.3 to +40	V
Switch Collector Voltage	$V_{C(switch)}$	40	V
Switch Emitter Voltage ($V_{Pin\ 1}$ = 40 V)	$V_{E(switch)}$	40	V
Switch Collector to Emitter Voltage	$V_{CE(switch)}$	40	V
Driver Collector Voltage	$V_{C(driver)}$	40	V
Driver Collector Current *	$I_{C(driver)}$	100	mA
Switch Current	I_{SW}	1.2	A
Power Dissipation and Thermal Characteristics T_A = 25℃	P_D	625	mW
Thermal Resistance	$R_{\theta JA}$	160	℃/W
Operating Junction Temperature	T_J	150	℃
Operating Ambient Temperature Range	T_A	0 to +70	℃
Storage Temperature Range	T_{stg}	-65 to +150	℃

* Maximum package power dissipation limits must be observed.

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■ Electrical Characteristics Ta = 25℃

Parameter	Symbol	Testconditons	Min	Typ	Max	Unit
Frequency	f_{osc}	$V_{Pin\ 5}$ = 0 V, C_T = 1.0 nF, T_A = 25℃	24	33	42	kHz
Charge Current	I_{chg}	V_{CC} = 5.0 V to 40 V, T_A = 25℃	24	35	42	μA
Discharge Current	I_{dischg}	V_{CC} = 5.0 V to 40 V, T_A = 25℃	140	220	260	μA
Discharge to Charge Current Ratio	I_{dischg}/I_{chg}	Pin 7 to V_{CC}, T_A = 25℃	5.2	6.5	7.5	
Current Limit Sense Voltage	$V_{ipk(sense)}$	Ichg = Idischg, T_A = 25℃	250	300	350	mV
Saturation Voltage, Darlington Connection	$V_{CE(sat)}$	I_{SW} =0.8 A, $P_{ins\ 1}$, 8 connected		1.0	1.3	V
Saturation Voltage (Forced β ≈ 20)	$V_{CE(sat)}$	I_{SW} =0.8A, $R_{Pin\ 8}$ = 82 Ω to V_{CC},		0.45	0.7	V
DC Current Gain	h_{FE}	I_{SW} =0.8 A, V_{CE} = 5.0 V, T_A = 25℃	50	75		
Collector Off-State Current	$I_{C(off)}$	V_{CE} = 40 V		0.01	100	μA
Threshold Voltage T_A = 25℃ T_A =0℃ to 70℃	V_{th}		 1.225 1.21	 1.25 	 1.275 1.29	V
Threshold Voltage Line Regulation	R_{egline}	V_{CC} = 3.0 V to 40 V		1.4	5.0	mV
Input Bias Current	I_{IB}	Vin = 0 V		-20	-400	nA
Supply Current $V_{Pin\ 5}$ > V_{th}, Pin 2 = GND, remaining pins open	I_{CC}	V_{CC} = 5.0 V to 40 V, C_T = 1.0 nF, Pin 7 = V_{CC},			4.0	mA